Name:		Roll No.:		Date:	

Practical 1

Generate a First RetailPulse Report

Related topic: Introduction to IBM Cognos Analytics - Reporting

Objective

Navigate the Cognos Analytics Reporting environment and generate a running list report against the RetailPulse Sales package.

Materials required

IBM Cognos Analytics access, RetailPulse Sales package, personal content folder (My content).

Procedure

1. Sign in to Cognos Analytics. On the welcome screen click **New (+)** in the left navigation bar, then choose **Report**.
2. In the *Select a package or data module* dialog, open **Team content** (or search) and select the **RetailPulse Sales** package. Click **OK**.
3. In the template gallery choose **List** (a blank report with a single list is created on Page 1).
4. Open the **Source** tab (Insertable objects). Expand the package and drag **Region**, **Product Category** and **Sales Amount** into the list on the canvas. Each becomes a column in that order.
5. Click the **Run** (play) button in the toolbar and select **Run HTML**. Live RetailPulse rows appear in a new tab. Close it to return to authoring.
6. Click **Save as**, browse to **My content**, enter the name **RetailPulse – First Sales Report** and click **Save**.
7. Close the report, then reopen it from My content to confirm it still returns live data.

[Paste your Cognos screenshot here]
Fig 1.1 - List report on canvas (authoring view)

Illustrative output (Run HTML)

Region	Product Category	Sales Amount
North	Apparel	$8,559.09
North	Electronics	$24,960.05
North	Grocery	$1,901.89
South	Apparel	$5,805.32
South	Electronics	$20,197.57
South	Grocery	$1,566.68
West	Apparel	$5,910.95
West	Electronics	$25,437.06
West	Grocery	$1,384.29

Columns are labelled with the package item names; every row is data returned by the query, not placeholder text.

[Paste your Cognos screenshot here]
Fig 1.2 - Run output

Verification checklist

[ ] Canvas shows 'Region / Product Category / Sales Amount' headers with data-item placeholders (not 'Text').

[ ] Run output shows real rows and no error banner.

[ ] Report saved as 'RetailPulse – First Sales Report' in My content.

[ ] Reopening and re-running returns live data.

Result

A three-column running list report was created from the RetailPulse Sales package, executed successfully against live data, saved in the personal folder and reopened correctly.

Name:		Roll No.:		Date:	

Practical 2

Build a Grouped RetailPulse Sales List Report

Related topic: Create List Reports

Objective

Construct a list report, group it by Region and Store Name, format columns and add a summarizing footer and page header.

Procedure

1. **New > Report**, select RetailPulse Sales, choose the **List** template (or Save-as your Practical 1 report).
2. Drag **Region, Store Name, Product Category, Quantity, Sales Amount** onto the list in that order.
3. Select the **Region** column, click **Group/Ungroup** in the toolbar. Repeat for **Store Name**. Order matters: Region first, Store Name second. Duplicate values collapse into group bands.
4. Select **Sales Amount** (and Quantity). In the Properties pane set **Horizontal alignment = Right**. For Sales Amount open **Data Format** > Format type **Currency**, 2 decimals.
5. With Sales Amount selected click **Summarize > Total**. Cognos adds a footer total per Store Name group and a grand total at the end (use Aggregate = Total / sum).
6. Switch to **Page layout**. In the page header insert a **Text item** 'RetailPulse – Grouped Sales Report' and from **Toolbox** insert a **Date** item (Date style: today's run date).
7. Run as HTML (and PDF to verify the header repeats on every page), then Save as **RetailPulse – Grouped Sales Report**.

Expression / configuration

```
Group order    : Region > Store Name
Sales Amount   : Currency, 2 dp, right aligned, Summarize = Total
Page header    : Text 'RetailPulse – Grouped Sales Report' + Date (today)
Alt. header calc: date2string(_today())  (layout calculation)
```

[Paste your Cognos screenshot here]
Fig 2.1 - Grouping and footer in authoring view

Illustrative output (first page header: RetailPulse – Grouped Sales Report | Run date: 09 Oct 2026)

Region	Store Name	Product Category	Quantity	Sales Amount
North	North Plaza	Apparel	101	$4,101.76
		Electronics	87	$14,184.70
		Grocery	85	$825.03
	Total - North Plaza		**273**	**$19,111.49**
	Northgate Mall	Apparel	103	$4,457.33
		Electronics	71	$10,775.35
		Grocery	113	$1,076.86
	Total - Northgate Mall		**287**	**$16,309.54**
South	Southside Central	Apparel	76	$3,116.27
		Electronics	67	$11,935.33
		Grocery	72	$666.60

Region	Store Name	Product Category	Quantity	Sales Amount
	Total - Southside Central		**215**	**$15,718.20**
	Harbor Point	Apparel	66	$2,689.05
		Electronics	52	$8,262.24
		Grocery	102	$900.08
	Total - Harbor Point		**220**	**$11,851.37**
West	Westfield Square	Apparel	78	$2,981.91
		Electronics	50	$7,645.08
		Grocery	59	$567.53
	Total - Westfield Square		**187**	**$11,194.52**
	Sunset Outlet	Apparel	69	$2,929.04
		Electronics	104	$17,791.98
		Grocery	85	$816.76
	Total - Sunset Outlet		**258**	**$21,537.78**
Grand total			**1440**	**$95,722.90**

[Paste your Cognos screenshot here]
Fig 2.2 - Run output with footers

Verification checklist

[ ] Region bands contain nested Store Name bands; repeated names are collapsed.

[ ] Sales Amount shows $ and 2 decimals; Quantity and Sales Amount right-aligned.

[ ] Footer total under each Store Name equals manual sum of its rows.

[ ] Page header with title and date appears on every page (check PDF output).

Result

A two-level grouped list report with currency formatting, per-store totals and a dated page header was built, run and saved.

Name:		Roll No.:		Date:	

Practical 3

Filter RetailPulse Sales by Region and Date

Related topic: Focus Reports Using Filters

Objective

Apply a simple filter and an advanced detail filter to scope a report to one region and the current quarter, then generalize Region with a prompt.

Procedure

1. Open **RetailPulse – Grouped Sales Report** from My content (Edit mode). Note the unfiltered row count.
2. Open the **Filters** pane (toolbar funnel icon) and choose **Add**. Select **Region**, operator **Equal to**, value **North**. Click **OK**.
3. Click the report's list, then **Filters > Create advanced filter > Detail filter** (applied to individual rows before summarizing). Enter the date expression below.
4. Make sure the two filters are joined with **AND** (both are listed as separate detail filters; Cognos ANDs them by default). Set Usage = **Required**.
5. Run the report. Only North rows dated within the current quarter remain, and the grand total drops.
6. To prompt: edit the Region filter, change the value choice to **Prompt every time** (or set the expression to the parameter below). Cognos generates a prompt page. Run and choose **South** to confirm different data returns.
7. Save as **RetailPulse – Filtered Sales Report**.

Expression / configuration

```
Simple filter   : [RetailPulse Sales].[Region] = 'North'
Detail filter   : [RetailPulse Sales].[Sales Date] between 2026-10-01 and 2026-12-31
Combined (AND)  : [Region] = 'North' AND [Sales Date] between 2026-10-01 and 2026-12-31
Prompted version: [Region] = ?pRegion?
```

[Paste your Cognos screenshot here]
Fig 3.1 - Filters pane with both conditions

Illustrative output - row counts before and after filtering

Scenario	Detail rows	Quantity	Sales Amount
Unfiltered (all regions, all dates)	36	1440	$95,722.90
Region = North only	12	560	$35,421.03
North AND Q4 2026 (1 Oct - 31 Dec)	6	269	$13,174.49
Prompt = South, Q4 2026	6	184	$10,518.08

Detail rows = underlying summary records in the sample data; in your package the counts will differ but must shrink in the same order.

[Paste your Cognos screenshot here]
Fig 3.2 - Filtered output

Verification checklist

[ ] Only North rows appear; all dates fall in the quarter.

[ ] Filters are combined with AND (not OR).

[ ] Row count / totals are visibly lower than the unfiltered report.

[ ] Re-run with South via the prompt returns different data.

Result

The report was scoped to North-region, current-quarter data using a simple and a detail filter joined by AND, and Region was converted to a reusable prompt.

Name:		Roll No.:		Date:	

Practical 4

Build a RetailPulse Region × Product Crosstab

Related topic: Create Crosstab Reports

Objective

Summarize sales across Region and Product Category with two measures and grand totals.

Procedure

1. **New > Report**, select RetailPulse Sales, choose the **Crosstab** template.
2. Drag **Region** to the **Rows** drop zone and **Product Category** to the **Columns** drop zone.
3. Drag **Sales Amount** into the **Measure** (body) cells.
4. Drag **Quantity** next to Sales Amount in the column edge (nested under each category) so both measures show per category.
5. Select the Region row header, **Summarize > Total** for the column totals; select the Product Category header, **Summarize > Total** for the row totals. Both axes now show grand totals.
6. Select Sales Amount, Data Format > **Currency**. Run as HTML; check alignment; Save as **RetailPulse Region-Product Crosstab**.
7. Reconcile: the grand total must equal the Practical 2 grand total ($95,722.90).

[Paste your Cognos screenshot here]
Fig 4.1 - Crosstab authoring view

Illustrative output

Region	Apparel		Electronics		Grocery		Total	
	Sales	Qty	Sales	Qty	Sales	Qty	Sales	Qty
North	$8,559.09	204	$24,960.05	158	$1,901.89	198	$35,421.03	560
South	$5,805.32	142	$20,197.57	119	$1,566.68	174	$27,569.57	435
West	$5,910.95	147	$25,437.06	154	$1,384.29	144	$32,732.30	445
Total	**$20,275.36**	**493**	**$70,594.68**	**431**	**$4,852.86**	**516**	**$95,722.90**	**1440**

[Paste your Cognos screenshot here]
Fig 4.2 - Run output

Verification checklist

[ ] Regions down the side, categories across the top, no misaligned cells.

[ ] Each cell shows summed Sales Amount and Quantity.

[ ] Row totals and column totals sum correctly; grand total matches Practical 2.

[ ] Sales Amount is currency-formatted everywhere.

Result

A Region × Product Category crosstab with two measures and row/column grand totals was built and its totals reconciled with the grouped list report.

Name:		Roll No.:		Date:	

Practical 5

Build the RetailPulse Sales Prompt Report

Related topic: Design a Cognos Prompt Report for Sales Analysis

Objective

Create a prompt page with a value prompt, a date-range prompt and a cascading Store prompt, each linked to a filter on the crosstab page.

Procedure

1. Open **RetailPulse Region-Product Crosstab** in Edit mode and Save as **RetailPulse Sales Prompt Report**.
2. Open **Pages** > **Prompt pages** > **New prompt page** (or Page explorer > Add page > Prompt page). Drag it above the crosstab page.
3. From **Toolbox** drag a **Value prompt**. In the wizard set Parameter name **pRegion**, choose **Region** as the data item. Set Required = Yes.
4. Drag a **Date prompt** (range) / two Date prompts for **pStartDate** and **pEndDate**, or use a *Date & time prompt* with Range = Yes. Set defaults (e.g. 2026-10-01 to 2026-12-31).
5. Drag a second **Value prompt**, parameter **pStore**, data item **Store Name**. Select it, set **Cascade source** = **pRegion**. Cognos now lists only stores in the chosen region.
6. On the crosstab page open Filters > Add detail filters using the parameters below (this links each control to a filter).
7. Select the Region prompt and set **Default selections** to North. Insert a **Finish** button (Toolbox > Prompt buttons) if not present.
8. Run. Test: North + current quarter, then change to South and verify the Store list and crosstab update.

Expression / configuration

```
[Region]     = ?pRegion?
[Store Name] in ?pStore?
[Sales Date] between ?pStartDate? and ?pEndDate?
Store prompt : Cascade source = pRegion
Defaults     : Region=North, 2026-10-01 to 2026-12-31
```

[Paste your Cognos screenshot here]
Fig 5.1 - Prompt page in authoring view

Illustrative prompt page and result

Prompt control	Type	Parameter	Test value
Region	Value prompt (list)	pRegion	North
Reporting period	Date range	pStartDate / pEndDate	2026-10-01 to 2026-12-31
Store	Value prompt, cascades from Region	pStore	North Plaza, Northgate Mall (only North stores listed)

Region = North | Q4 2026	Apparel	Electronics	Grocery	Total
Sales Amount	$4,632.58	$7,492.12	$1,049.79	$13,174.49

Region = South | Q4 2026	Apparel	Electronics	Grocery	Total
Sales Amount	$2,391.74	$7,396.93	$729.41	$10,518.08

Changing Region from North to South changes the Store list (Southside Central, Harbor Point) and the crosstab figures.

[Paste your Cognos screenshot here]
Fig 5.2 - Prompt page run view and filtered crosstab

Verification checklist

[ ] Prompt page shows Region, date range and Store controls.

[ ] Changing Region refreshes the Store list without error.

[ ] Submitting returns a crosstab limited to the selected values.

[ ] Defaults give a sensible report when Finish is clicked straight away.

Result

A prompt page with value, date-range and cascading prompts was added in front of the crosstab, all linked to filters, and tested end to end.

Name:		Roll No.:		Date:	

Practical 6

Build the RetailPulse Customer Insights Dashboard

Related topic: Build a Dashboard for Customer Insights Visualization

Objective

Assemble a dashboard with a summary card, trend line, segment bar chart and a shared Region filter for RetailPulse customer data.

Procedure

1. **New > Dashboard**. Choose a template (e.g. blank canvas / grid), then **Select a source** > **RetailPulse Customers** data module > **Add**.
2. **Summary card:** Add a **Summary** visualization (KPI). Drag **Customer ID** (or Active Customers measure) to Value; set aggregation **Count distinct** and, if provided, filter **Status = Active**. Title: 'Total Active Customers'.
3. **Trend:** Add a **Line** visualization. X axis: **Month** (date). Y axis: **New Customers** and **Returning Customers** (or Customer Type as the color field). Title: 'New vs Returning Customers'.
4. **Segment bar:** Add a **Column/Bar** chart. Category: **Loyalty Segment**; Value: customer count. Sort descending by value.
5. **Shared filter:** Add a **Filter** widget (Add > Filters) or drag **Region** to the dashboard filter bar. Choose **Apply to all visualizations** (scope: whole dashboard). Select North and verify all three widgets change.
6. **Layout:** Resize and place the KPI at top-left, filter at top, trend line wide in the middle, segment bar below or beside it. Keep alignment to the grid, no overlaps; add titles.
7. **Save** as **RetailPulse Customer Insights Dashboard** in My content, close and reopen to confirm.

[Paste your Cognos screenshot here]
Fig 6.1 - Dashboard authoring canvas

Illustrative dashboard (generated sample data)

View A - Region filter: All

RetailPulse Customer Insights

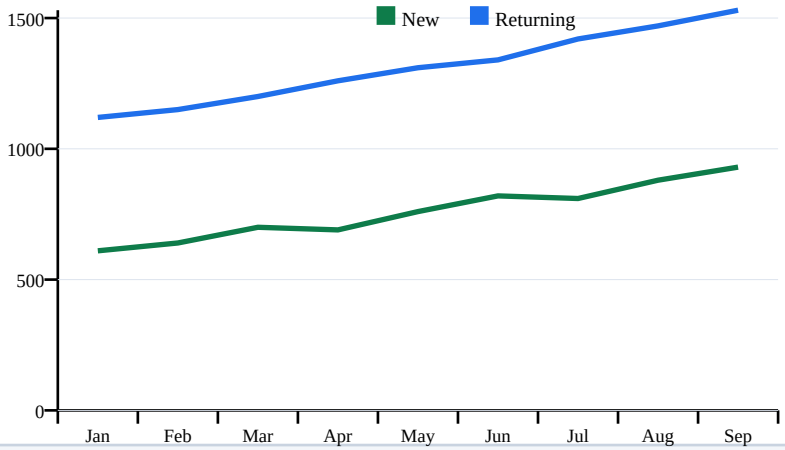
Region: All

Total Active Customers

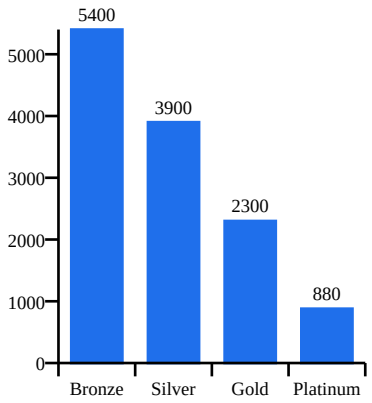
12,480

Selected period: Jan - Sep 2026

New vs Returning Customers



Customers by Loyalty Segment



View B - Region filter: North (all three widgets update; North sample = 34% of customers)

RetailPulse Customer Insights

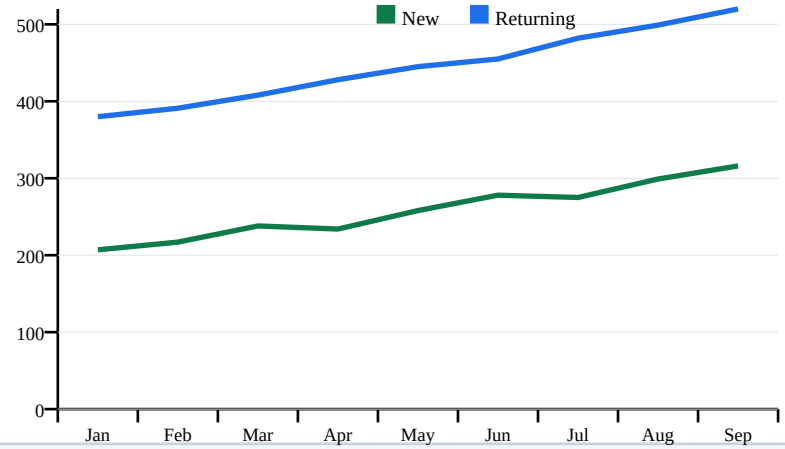
Region: North

Total Active Customers

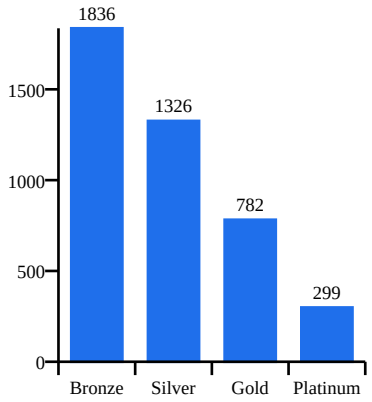
4,243

Selected period: Jan - Sep 2026

New vs Returning Customers



Customers by Loyalty Segment



[Paste your Cognos screenshot here]
Fig 6.2 - Your finished dashboard

Verification checklist

[ ] Summary card, trend line and segment bar all show live data (no placeholders).

[ ] Selecting a region in the filter widget updates every widget consistently.

[ ] Trend line renders the whole period with a legend.

[ ] Layout has a clear hierarchy (KPI/filter on top, charts below) with no overlaps.

[ ] Dashboard saved and reopens correctly.

Result

A customer-insights dashboard with a KPI card, new-vs-returning line chart, loyalty-segment bar chart and a global Region filter was built and saved.

Troubleshooting quick reference

Empty report: check package security / date filter range. **Totals look doubled:** a column is not summarized correctly or a join duplicates rows - check aggregate = Total. **Cascade not working:** Cascade source must be the *parameter name* of the Region prompt (pRegion). **Filter widget does not affect a chart:** set filter scope to Dashboard (all visualizations) and confirm all widgets use the same data module.